

Author Network Dynamics in 20th Century Anglophone Philosophy

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How can computational approaches serve the history of philosophy?

- Academic philosophy becomes increasingly professionalized over the course of the 20th century
- History of philosophy is often centred around towering figures
- Data-driven analyses promises to complement such approaches because they allow a focus on
 - (1) minor, less recognized figures and
 - (2) general tendencies among a broader group of researchers in the field (specialization, division of labour, etc.)

Research Questions

Is there a decrease in attention to historical figures?

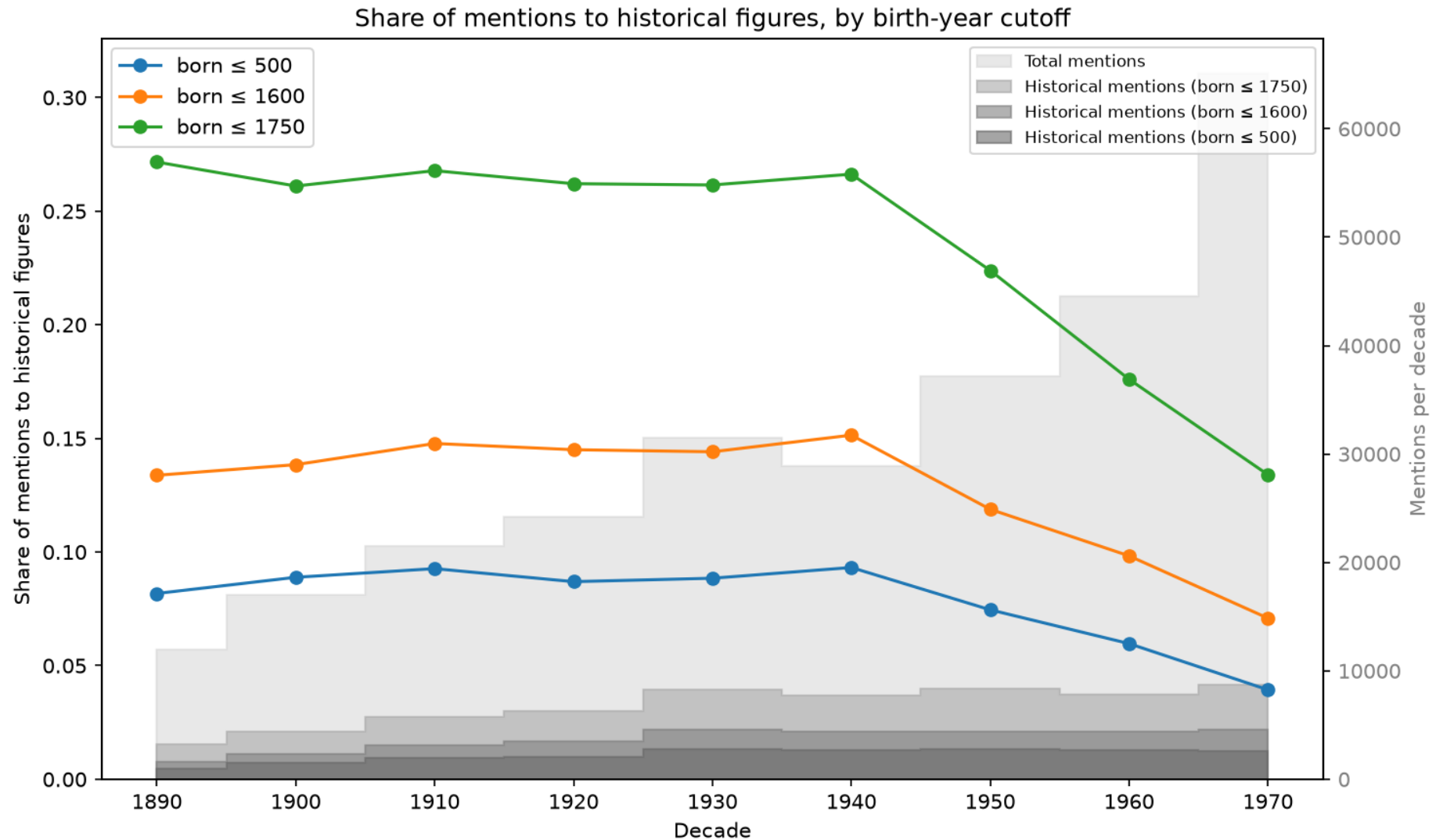
Is the specialization of philosophy reflected in the topical distributions of philosophers? Are minor figures more specialized?

How do communities of active philosophers develop over time?

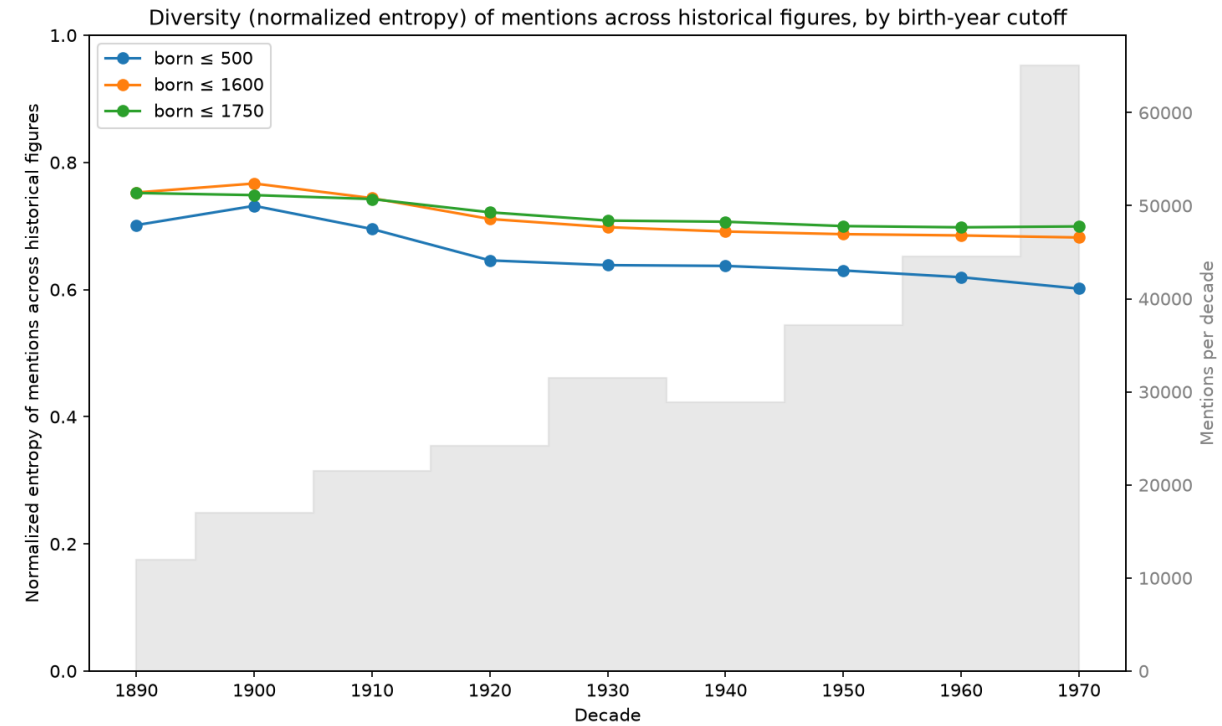
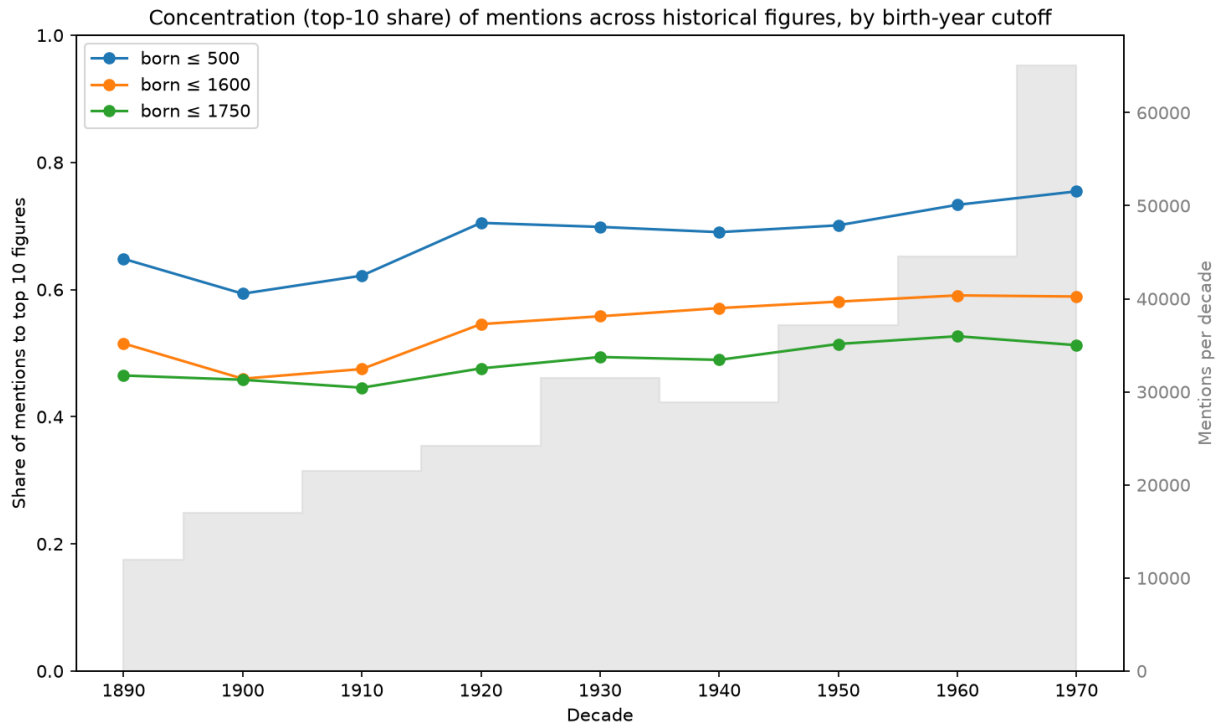
Data: constructing edhiphy

- Corpus: 22,977 article full texts (JSTOR OCR) from 12 anglophone philosophy journals between 1890 and 1980
- Metadata for 44,376 philosophers,
including partial PhD and affiliation data
Aggregated from WikiData, JSTOR, ProQuest, and manual addition
- Named Entity Recognition with a Gazetteer (for recall and transparency)
- Deterministically disambiguate >1,000,000 mentions,
ca. 87% accuracy on validation sample of 1,000
- Interactive exploration at <https://edhiphy.org>

Attention to Historical Figures



Attention Inequality



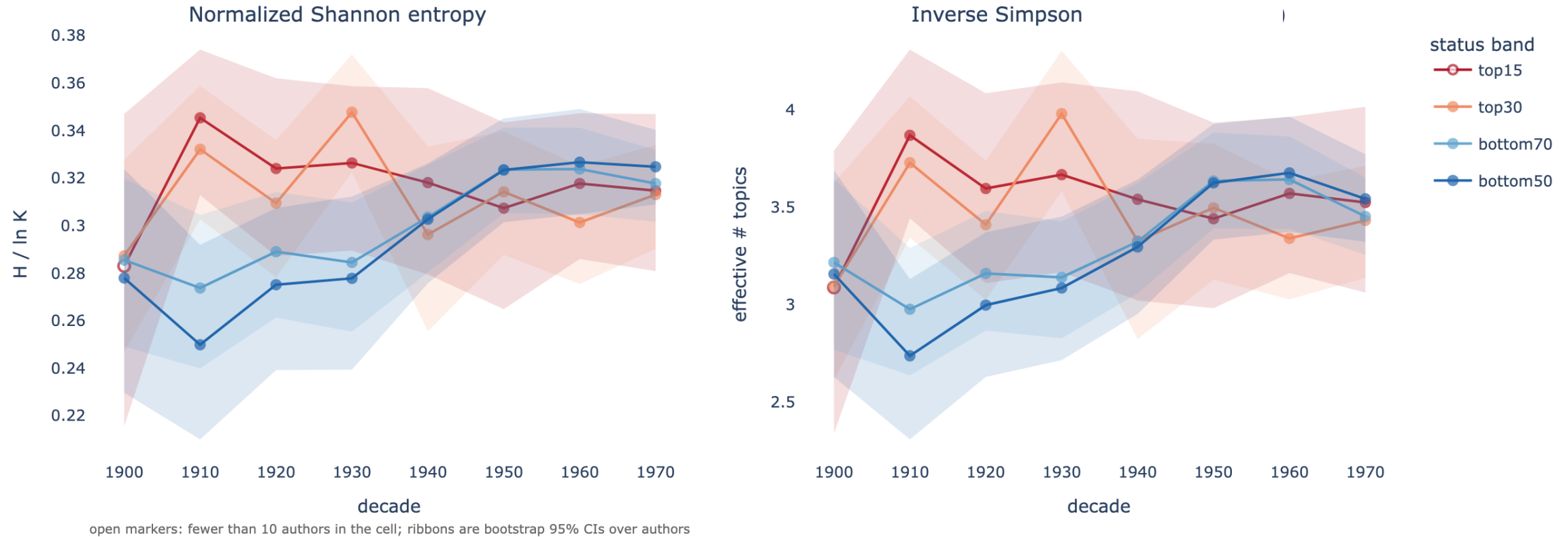
Topic modelling via STM

- Articles are not divided into paragraphs uniformly across corpus, which is why we choose an STM (Roberts, 2018) instead of a transformer-based model such as BERTopic
- Excluded philosopher mentions, stop words and non-English articles from corpus
- Preliminary choice: 60 topics, LLM-labelled
- Allows investigation of diversity measures based on content (and not only on referenced philosophers)

Topic diversity over time: Comparison

Can we see the specialization of philosophy in terms of topics, and how does it differ between minor and major figures?

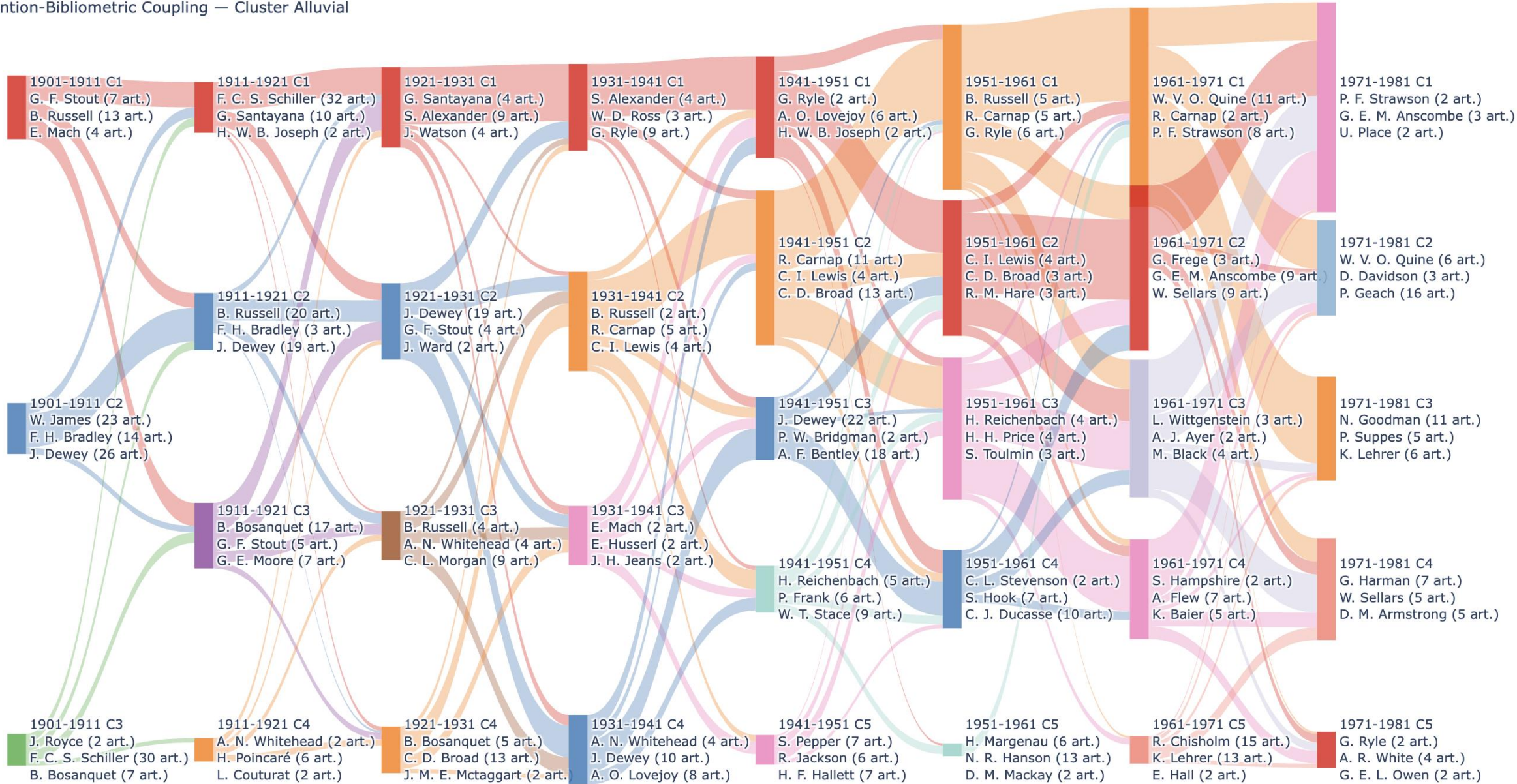
Topic diversity by status band — fixed 5 articles per author-decade (K=60, 4 topics dropped)

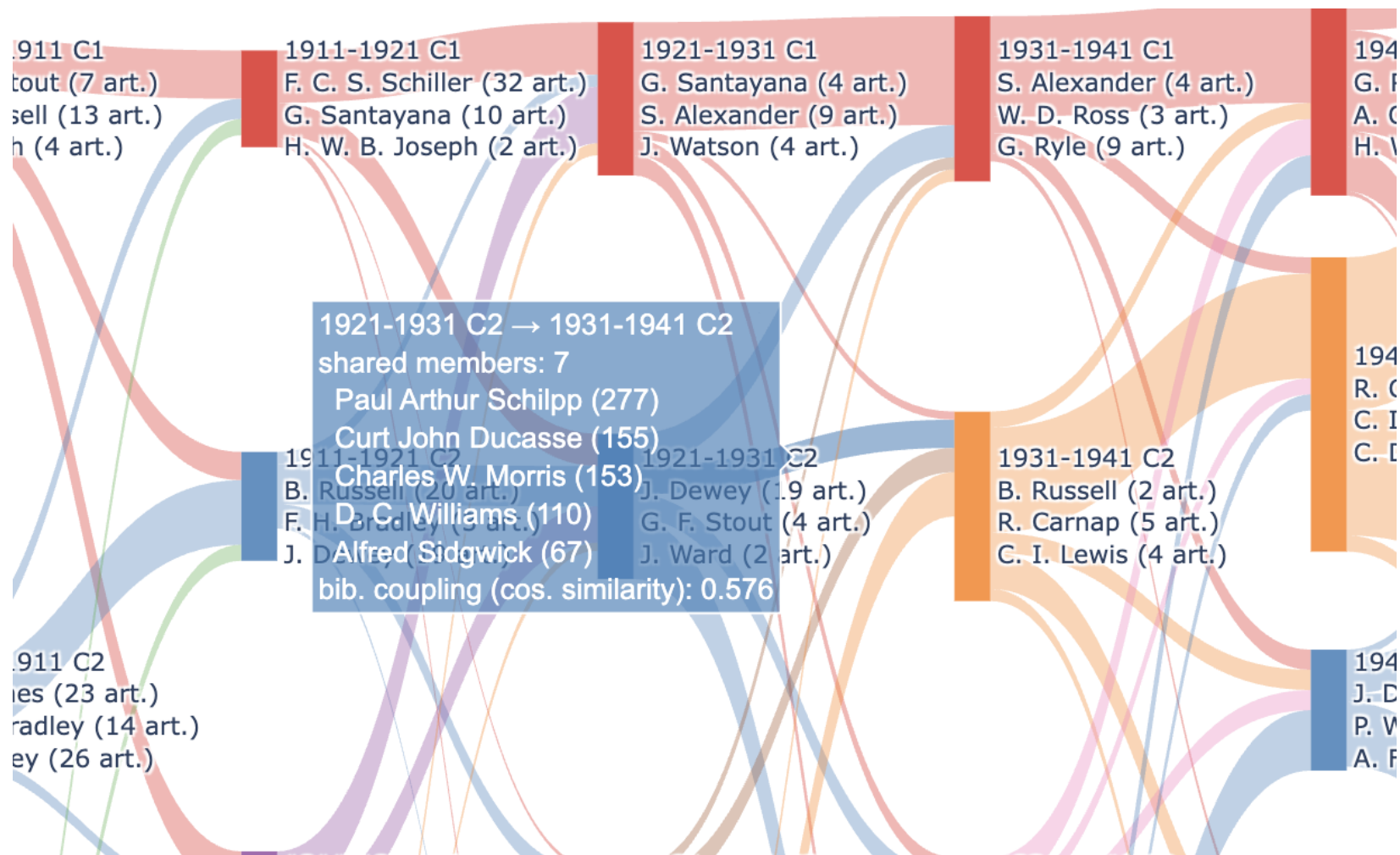


Mention-/author-bibliographic coupling networks

- "[M]ethod to map the research activities of active authors themselves for a more realistic picture of the current state of research in a field" (Zhao & Strotmann, 2008)
- Nodes: active authors with at least 2 publications in the observation period
- Links: pairwise cosine similarity of the vector of mentioned philosophers
- **Dynamic approach: Decade-wise modularity-based clustering of each network (Louvain)**
- Communities identified over time via maximum cosine similarity of their collective mention vector with cutoff

Mention-Bibliometric Coupling — Cluster Alluvial





Conclusion

- Preliminary results:
 - Attention to historical figures decreases (number of mentions stays relatively constant, but overall mentions increase)
 - Topic diversity decreases for major figures and increases for minor figures over time
 - Alluvial diagram: allows to trace participant migration among main communities, but high complexity

Open Questions

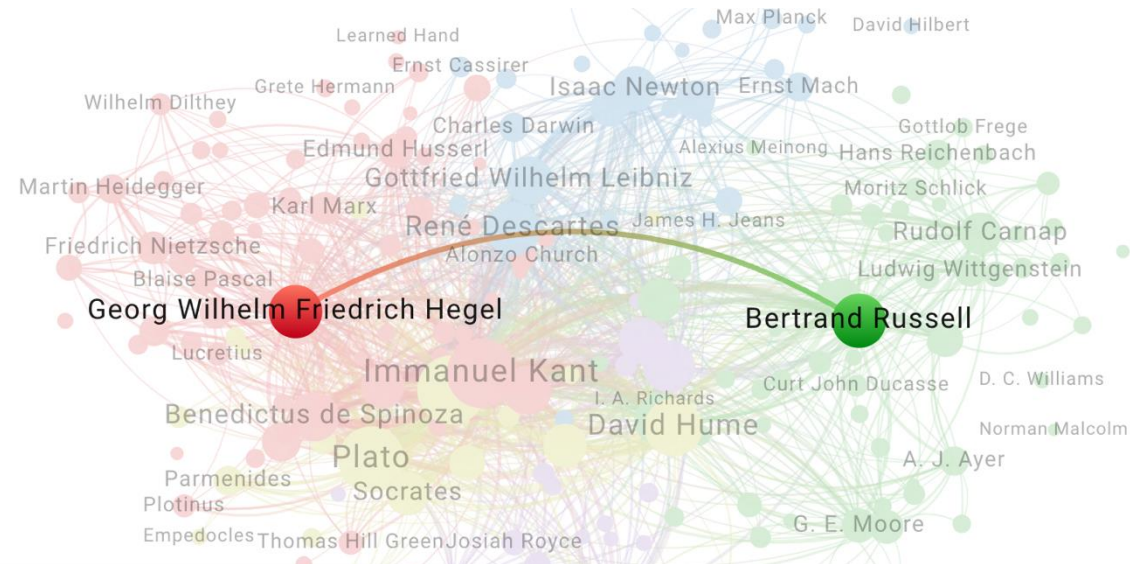
- How can we operationalize 'specialization'? (Problem: topic number increases over time which naturally increases entropy/diversity --> distance measure between topics?)
- Mention-bibliographic network: Problem of identifying communities over time; how can we move from snapshots to real network dynamics?
- All philosophers in the data set are somewhat influential figures
- Can the anglophone dynamics be contrasted with a francophone corpus?

Acknowledgements

edhiphy

Explore ▾

Visualize ▾



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- References:
 - Zhao, D. and Strotmann, A. (2008) 'Evolution of research activities and intellectual influences in information science 1996–2005: Introducing author bibliographic-coupling analysis', *Journal of the American Society for Information Science and Technology*, 59(13), pp. 2070–2086. Available at: <https://doi.org/10.1002/asi.20910>.
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